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CV-22 EVALUATION CRITERIA

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This manual implements AFPD 11-2, *Aircraft Rules and Procedures,* and AFPD 11-4, *Aviation Service*. This is a specialized publication intended for use by Airmen who have graduated from technical training related to this publication. It establishes evaluation criteria for initial and periodic aircrew qualification for all CV-22 units. This publication applies to the Regular Air Force and Air National Guard. This publication does not apply to Air Force Reserve Command units. It is used in conjunction with Air Force Instruction (AFI) 11-202V2, *Aircrew Standardization/Evaluation Program*, and Major Command (MAJCOM) supplements thereto. The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. Submit requests for waivers through the chain of command to the appropriate tier waiver authority, to the requestor's commander for non-tiered compliance items. See AFI 33-360, *Publications and Forms Management*, for a description of the authorities associated with the Tier numbers. This publication may be supplemented at any level, but all Supplements must be routed to the OPR of this publication for coordination prior to certification and approval. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the AF Form 847, *Recommendation for Change of Publication*; route AF Forms 847 from the field through the appropriate functional chain of command. Ensure all records created as a result of processes prescribed in this publication are maintained in accordance with Air Force Manual 33-363, *Management of Records*, and disposed of in accordance with the Air Force Records Disposition Schedule located in the Air Force Records Information Management System. This manual requires the collection and or maintenance of information protected by the Privacy Act of 1974 authorized by Title 37 United States Code 301a *Incentive Pay*, Public Law 92-570, *Appropriations*

Act of 1973, PL 93-294, *Aviation Career Incentive* 1974, Department of Defense Directive 7730.57, *Aviation Career Incentive Act of 1974 and Required Annual Report, February 5, 1976, with Changes 1 and 2,* and Executive Order 13478, *Amendments to Executive Order 9397 Relating to Federal Agency Use of Social Security Numbers.* The applicable System of Records Notice F011 AF XO A, Aviation Resource Management System (ARMS) is available at: **http://dpclo.defense.gov/Privacy/SORNs.aspx**.

SUMMARY OF CHANGES

This document has been substantially revised and must be thoroughly reviewed. Major changes include clarifying evaluator requirements during initial qualification, clarifying information on consecutive Instrument/Qual/Instructor Mission evaluations utilizing Aircrew Training Devices (ATDs), removal of Classified Material/Operations Security and Anti-Hijacking/Aircraft Security from **Table 2.1**, Instructor Mission evaluations utilizing ATDs, changes to Night Vision Google (NVG) Low-Level, NVG Tactical Approval and Landing, Low Visibility Approach, Time-on-Target and Alternate Insertion/Extractions Q/Q- standards, significant changes to **Table 5.1**, and removal of **Chapter 6** Direct Support Operator (DSO) Evaluations. Waiver authority guidance has been revised to comply with Tier Waiver Authority guidance.

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Chapter 1

GENERAL INFORMATION

1.1. General. This manual establishes requirements and grading criteria for ground and flight phases of initial, requalification and periodic flight evaluations. Aircrew evaluations will be conducted in accordance with (IAW) this manual and AFI 11-202V2, as supplemented. **(T-2).** Specific areas for evaluation are prescribed to ensure an accurate assessment of the proficiency and capabilities of aircrews. Flight examiners use this AFI when conducting aircrew evaluations. Instructors use this AFI when preparing aircrews for qualification.

1.2. Key words and definitions.

1.2.1. "Will" and "Shall" indicate a mandatory requirement.

1.2.2. "Should" is normally used to indicate a preferred, but not mandatory, method of accomplishment.

1.2.3. "May" indicates an acceptable or suggested means of accomplishment.

1.2.4. "Note" indicates operations procedures, techniques, etc., considered essential to emphasize.

1.3. Waivers. Waiver authority for this publication is AFSOC/A3, IAW AFMAN 11-202, Vol 2. Do not deviate from the policies and guidance in this AFMAN, except for safety or when necessary to protect the crew or aircraft from a situation not covered by this AFMAN and immediate action is required. Report deviations or exceptions without waiver through channels to MAJCOM Standardization/Evaluation function who in turn, will notify AFSOC/A3V for follow-on action, if necessary (**T-3**). Submit requests for waivers through the chain of command to the appropriate tier waiver approval authority, or alternately, to the requestors commander for non-tiered compliance items.

1.4. Revisions. Personnel are encouraged to submit proposed procedure changes through stan/eval channels IAW AFI 11-215.

1.5. Supplements. Units are encouraged to supplement this manual with standard evaluation profiles that best fit the unit's mission, equipment, and location.

1.6. Evaluation Procedures. Flight examiners will use the evaluation criteria in this volume to conduct all flight, aircrew training device (ATD), and emergency procedure evaluations (EPE). **(T-2).** Flight examiners will brief the examinee on the conduct, purpose, and requirements of the evaluation, as well as all applicable evaluation criteria prior to flight. **(T-3).** Flight examiners will ensure the examinee or appropriate team crew member accomplishes all required mission planning. **(T-3).** If an Operations Planning Team or Deployment Planning Team accomplishes mission planning, the examinee is ultimately responsible for the accuracy and completeness of all mission-planning paperwork. Flight examiners will be furnished a copy of necessary charts, flight logs, mission folders, and any additional items they deem necessary. **(T-3).**

1.7. Roles and Responsibilities. Flight examiner and examinee responsibilities.

1.7.1. Flight examiners will ensure all required training and documentation are complete prior to initial/requalification evaluations. **(T-3).** Flight examiners will debrief/critique all aspects of the flight. (**T-3).** During the critique, the flight examiner will review the examinee's overall

rating, specific deviations, area/sub area grades assigned, and any additional training required. **(T-3).**

1.7.2. Pilot flight examiners should occupy a primary crew position during evaluations. Flight Engineer examiners will not occupy a primary crew position while administering evaluations. **(T-3).**

1.7.3. Flight examiners will not intentionally fail any equipment during flight evaluations, but may deny the use of systems not affecting safety of flight. **(T-3).** (**Exception**: During evaluations conducted in ATDs, equipment may be failed or disabled.)

1.7.4. During pilot initial or requalification evaluations in the aircraft either a qualified instructor pilot or flight examiner will be at a set of controls. **(T-2).** (**Exception**: The FTU may conduct initial Instrument Evaluations in the ATD with two unqualified pilots at the controls.)

1.7.5. Examinee will be current for all events evaluated during a periodic evaluation. **(T-3).** Unless requested by examinee and approved by squadron operations officer or designated representative.

1.8. Grading Instructions. To ensure standard and objective evaluations, flight examiners will be thoroughly familiar with the prescribed evaluation criteria. **(T-3).** Flight examiner judgment in applying the grading system prescribed by AFI 11-202V2, will be the determining factor in arriving at the overall grade. **(T-3).** The evaluator should base tolerances for in-flight parameters on conditions of smooth air and a stable aircraft. Do not consider a momentary deviation from tolerances provided the examinee applies prompt corrective action and such deviations do not jeopardize flying safety. The evaluator will consider cumulative deviations when determining the overall grade. **(T-3).**

1.9. Critical Areas. Critical areas require adequate accomplishment by the aircrew member in order to successfully achieve the mission objectives.

1.9.1. If an aircrew member receives an unqualified grade in any critical area, the overall grade for the evaluation will also be unqualified. **(T-2).**

1.9.2. Critical areas are identified by (**CRITICAL**) in the area title and by shading of the Q-block on the appropriate evaluation worksheet.

1.9.3. Safety/judgment, aircrew discipline, airmanship/situational awareness (SA), instructional abilities and boldface are considered critical. If one of these sub-areas is graded U, then the overall grade for the evaluation will be Q-3 as defined in AFI 11-202V2, paragraph 5.3.3.3.

1.10. Evaluation Requirements. Conduct flight evaluations as specified in AFI 11-202V2 as supplemented, the specific crew position chapter and associated tables of this volume. Examiners will use the criteria contained in this volume to accomplish all flight, ATD, and emergency procedures evaluations. Initial qualification (QUAL), Initial/Periodic Mission (MSN) and Special Mission Event (SME) evaluations will be accomplished in the aircraft. **(T-2). Exception**: Instructor Mission (INSTR MSN) evaluations may be conducted in the simulator per **paragraph 3.5**.

1.10.1. Consecutive Simulator Evaluations. Two consecutive Qualification (QUAL) evaluations should not normally be conducted in an ATD. However, based on operational

requirements, consecutive qualification evaluations may be conducted in an ATD with squadron commander approval annotated in the remarks section of the AF Form 8 *Certificate of Aircrew Qualification*. There is no restriction to consecutive instrument (INSTM) evaluations in an ATD.

1.10.2. Alternate Method. When it is impossible to evaluate an area in flight due to equipment malfunctions, operational requirements, scheduling restrictions, or weather, the area may be evaluated by an alternate method (i.e., ATD, or verbal examination). If, in the flight examiner's judgment, an item cannot be adequately evaluated by an alternate method, complete the evaluation on an additional flight.

1.10.3. If an area/subarea was not able to be evaluated in flight, and the event is certified for evaluation purposes in the simulator, it can be evaluated in the simulator to complete the evaluation. Document in the comments section of AF Form 8/8a, *Certificate of Aircrew Qualification,* which portion(s) of evaluation were conducted in simulator.

1.11. Unsatisfactory Performance. If the flight examiner observes an aircrew counterpart jeopardizing safety, the examiner will assume the duties of that aircrew member (provided the examiner's flight duty period (FDP) does not exceed AFMAN 11-202V3, *General Flight Rules* maximum FDP for an un-augmented crew). **(T-3).** This does not mean the examiner must assume the examinee's position any time unsatisfactory performance is observed. If the examiner feels the examinee can continue safely with supervision, the examiner is not required to assume the examinee's duties.

1.11.1. If the flight examiner assumes the examinee's duties, assign a Qualification Level 3 (Q-3) as the overall grade.

1.11.2. Evaluators must report deviations/discrepancies from established procedures/directives in any area, regardless of the individual's crew specialty, to squadron/group commander, along with evaluator's recommendation for corrective action, IAW AFMAN 11-202V2 and MAJCOM supplement. **(T-3).**

1.12. Additional Training. The decision to assign additional training is at flight examiner discretion, who must judiciously make the determination that additional training is needed. Document additional training and completion IAW AFI 11-202V2 and MAJCOM supplement.

1.12.1. Any approved training device or medium may be used for additional training.

1.12.2. Additional training may be accomplished on the same flight as the evaluation, provided the unique situation presents a valuable training opportunity and the discrepancy requiring the additional training will not result in overall Q-3 evaluation. **(T-3).** This option requires flight examiner discretion and judicious application. The examinee must be informed when the additional training begins and ends. **(T-3)**.

1.13. Rechecks. Rechecks, in order to remedy a failed evaluation, should normally be administered by a flight examiner other than the one who administered the original evaluation.

1.14. Flight Examiners.

1.14.1. Evaluators must be instructor qualified in a given event prior to acting as an evaluator in the event. **(T-2).**

1.14.2. Certified evaluators who subsequently add special mission instructor or other instructor qualifications are automatically certified to evaluate these new qualifications.

1.14.3. Evaluation that are administered by unit evaluators to unit members are designated as complete by either the unit chief of stan/eval flight examiner or the squadron commander. Evaluations that are administered by outside flight examiners (i.e. MAJCOM, cross command, Aircrew Stan/Eval Visit, etc.) are designated as complete by the evaluator.

1.15. Special Qualifications. Special qualification evaluations are administered for Special Mission Events (SME). These events are not universal to all crewmembers. Special qualification evaluations may be conducted separately or in conjunction with qualification/mission evaluations. After qualification, areas can be graded as part of periodic mission evaluations.

1.15.1. There are no requisites for special qualification evaluations (SME) unless specified. Refer to appropriate crew position chapter for any special qualification evaluation requirement, and AFMAN 11-202 Vol 2 for AF Form 8/8a procedures.

1.15.2. Special mission evaluations are defined per AFI 11-2CV-22V1 *CV-22, Aircrew Training* and this publication; and results in an AF Form 8/8a. Special mission evaluations will be annotated as SPOT evaluations in the Flight Phase block of the AF Form 8/8a and the Type of Evaluation block on the AF Form 942, *Record of Evaluation*. **(T-2).** If the SME status is "current/qualified" and accomplished during the evaluations profile, the SME will be evaluated during the MSN evaluation but annotated separately on the "Mission Description" section of the AF Form 8/8a. **(T-3).**

Chapter 2

ALL EVALUATIONS

2.1. General. The general grading criteria contained in this chapter applies to all crew positions and all evaluations. The examinee must satisfactorily demonstrate the ability to perform required duties safely and effectively. **(T-2).** This includes appropriate aircraft systems operation IAW applicable technical orders, instructions, and directives. Prerequisites are evaluation requirements that must be met before beginning an evaluation and requisites are evaluation requirements that must be completed before the evaluation is considered complete.

2.2. Requirements. Evaluate all crewmembers on areas listed in **Table 2.1**.

2.2.1. **Examinations.** All crew members will complete open and closed book examinations as a requisite to periodic evaluations IAW AFI 11-202V2 and applicable supplements. **(T-2).**

2.2.2. Emergency Procedures Evaluation (EPE). An EPE is a requisite for all QUAL and MSN evaluations except special mission evaluations. EPEs may be conducted verbally, in-flight, in a simulator, or by another method determined by the examiner or unit stan/eval. Operations Group stan/eval may develop EPE guides for each crew position for flight examiner use. EPEs should be scenario driven and tailored to the specific crew position. The EPE will include areas commensurate with the examinee's qualification and experience level. **(T-3).** Examiners should include other general knowledge areas as well. For mission evaluations, evaluate mission-specific equipment and situations. EPEs will include sufficient in-flight and ground emergencies to evaluate the examinee's knowledge of systems and procedures to the flight examiner's satisfaction. **(T-3).**

2.2.3. Examinees may use publications that are normally available inflight. The examinee must recite all boldface items from memory and should provide the initial steps of emergency procedures that, in the opinion of the examiner, would not allow time for reference. **(T-2).**

2.2.4. **Publications Check.** Required for all QUAL or combined QUAL/MSN evaluations as outlined in area 12 of **Table 2.1** General Grading Areas and AFI 11-215.

2.2.5. **Cockpit/Crew Resource Management (CRM).** In accordance with AFI 11-290, will be evaluated during initial and periodic evaluations. **(T-2).** CRM skills are integral to all phases of flight; therefore, no specific area titled CRM exists. CRM skills are imbedded within specific grading criteria (mission planning, airmanship/situational awareness, crew coordination, communication, risk management/decision making, task management, and briefing/debriefing) and include all of the skills listed on the AF Form 4031, *CRM Skills Criteria Training/ Evaluation Form*. Therefore, use of the AF Form 4031 is unnecessary for evaluations.

2.2.6. **Formal Course Evaluations.** All required areas must be evaluated for the type of evaluation flown, IAW guidance in this volume. **(T-2).** Training objectives and related areas should be graded using the performance criteria in this volume.

Table 2.1. General Grading Areas (all Crew Positions and all Evaluations).

Area	Notes	Grading Areas
1	1	Safety/Judgment – (**CRITICAL**)

Area	Notes	Grading Areas
2	1	Aircrew Discipline – (**CRITICAL)**
3	1	Airmanship/Situational Awareness – (**CRITICAL)**
4	2, 3	Boldface – (**CRITICAL)**
5	2	Emergency Procedures Evaluation
6	1	Crew Coordination
7	2	Mission Planning
8	2	Knowledge of Directives
9	1	Preflight
10	1	Use of Checklist
11	2	Forms/Reports/Logs
12	2	Personal/Professional Equipment/Flight Publications
13	2	Emergency and Life Support Equipment/Procedures
14	2	Briefings/Debriefings/Forms
15	1	Communication
16	2	Risk Management/Decision Making
17	1	Task Management

Notes:

1. Required in-flight or ATD.
2. Required in-flight or alternate method.
3. Only required for QUAL evaluation.

2.3. General Grading Criteria.

2.3.1. **Area 1. Safety/Judgment - (CRITICAL).**

2.3.1.1. Q. Was aware of, and complied with, all safety factors required for safe aircraft/ equipment operation and mission accomplishment. Identified and assessed risk appropriately. Properly considered consequences of decisions.

2.3.1.2. U. Not aware of, or did not comply with, all safety factors required for safe aircraft/ equipment operation or mission accomplishment. Failed to properly identify and assess risk. Failed to consider consequences of decisions. Operated the aircraft/equipment in a dangerous manner.

2.3.2. **Area 2. Aircrew Discipline - (CRITICAL).**

2.3.2.1. Q. Demonstrated strict professional flight and crew discipline throughout all phases of the mission.

2.3.2.2. U. Failed to exhibit strict flight and crew discipline. Violated or ignored rules or instructions.

2.3.3. **Area 3. Airmanship/Situational Awareness - (CRITICAL).**

2.3.3.1. Q. Executed the assigned mission in a timely, efficient manner. Anticipated situations that would have adversely affected the mission, and corrected them. Made

appropriate decisions based on available information. Recognized the need for action. Aware of performance of self and other flight members. Aware of on-going mission status. Recognized, verbalized and acted upon unexpected events.

2.3.3.2. U. Decisions or lack thereof caused failure to accomplish assigned mission. Did not recognize the need for action. Not aware of performance of self and other flight members. Not aware of on-going mission status. Failed to recognize, verbalize and act on unexpected events.

2.3.4. **Area 4. Boldface – (CRITICAL).**

2.3.4.1. Q. Able to recite/write the proper emergency boldface actions in the correct sequence with no discrepancies (not necessarily a verbatim response).

2.3.4.2. U. Failed to recite/write emergency boldface items in the correct sequence. Discrepancies in the procedure.

2.3.5. **Area 5. Emergency Procedures Evaluation.**

2.3.5.1. Q. Satisfactory systems/procedural knowledge. Operated within prescribed limits and correctly diagnosed problems. Performed and/or explained proper corrective action, in the proper sequence, for each type of malfunction. Accomplished all required checklists and/or effectively used available aids. Thoroughly described the location, use and limitations of emergency equipment.

2.3.5.2. Q-. Marginal systems/procedural knowledge. Slow to analyze problems or apply proper corrective actions. Did not effectively use checklist and/or available aids. Minor omissions or deviations in describing the location, use and limitations of emergency equipment.

2.3.5.3. U. Unsatisfactory systems/procedural knowledge. Failed to analyze problem or take corrective action. Failed to accomplish required checklists and/or unable to locate information in available aids. Major omissions or deviations in describing the location, use and limitations of emergency equipment.

2.3.6. **Area 6. Crew Coordination.**

2.3.6.1. Q. Provided direction/information when needed. Adapted to meet new situational demands and focused attention on the task. Knew assigned task of other crew members. Asked for inputs, and made positive statements to motivate crew members.

2.3.6.2. Q-. Crew coordination was limited though adequate to accomplish the mission. Provided limited direction/information when needed. Slow to adapt to meet new situational demands due to limited focus on task. Did not consistently seek inputs from other crew members. Limited effort to motivate crew members through positive statements.

2.3.6.3. U. Did not provide direction/information when needed. Did not adapt to meet new situational demands and focus attention on the task. Did not seek inputs or made no effort to make positive statements to motivate crew members. Lack of crew coordination resulted in significant degradation of mission accomplishment.

2.3.7. **Area 7. Mission Planning.**

2.3.7.1. Q. Clearly defined the mission overview and mission goals. Provided specific information on required tasks. Solicited feedback from other crew members to ensure understanding of mission requirements. Thoroughly critiqued plans to identify potential problem areas and ensured all had understanding of possible contingencies. Checked all factors applicable to flight such as Flight Information Publication, weather, Notices To Airman System, alternate airfields, flight logs, performance data, fuel requirements, and charts. When required, extracted necessary information from air tasking order/frag. Was aware of the available alternatives if unable to complete the flight/mission as planned. Read and initialed all items in the Flight Crew Information File/read files.

2.3.7.2. Q-. Did not adequately define the mission overview and mission goals. Potential problem areas partially addressed or not at all. Did not adequately solicit feedback or critique the plans to ensure understanding of possible contingencies. Minor errors or omissions detracted from mission effectiveness, but did not affect mission accomplishment. Limited knowledge of performance capabilities or approved operating procedures/rules.

2.3.7.3. U. Did not define the mission overview and goals. Lacked specific information on required tasks. Did not solicit feedback from other crew members to ensure understanding. Did not critique plans to identify potential problem areas. Major errors or omissions would have prevented a safe or effective mission. Unsatisfactory knowledge of operating data or procedures.

2.3.8. **Area 8. Knowledge of Directives.**

2.3.8.1. Q. Prepared and completed mission in compliance with existing instructions and directives. Demonstrated knowledge of operating procedures and restrictions and where to find them in the correct publications.

2.3.8.2. Q-. Minor deviations from procedures. Unsure of directives and/or had difficulty locating information in appropriate publications. Any instances of non-compliance did not jeopardize safety.

2.3.8.3. U. Unaware of procedures and/or could not locate them in the appropriate publication in a timely manner. Failed to comply with a procedure that could have jeopardized safety or mission success.

2.3.9. **Area 9. Preflight.**

2.3.9.1. Q. Completed systems preflight/inspections IAW tech orders, checklists, and instructions. Individual technique complied with established procedures.

2.3.9.2. Q-. Minor deviations from established systems pre-flight/inspection. Individual technique was safe, but detracted from established procedures. Used individual technique instead of established procedure and was unaware of differences.

2.3.9.3. U. Failed to preflight critical component or could not conduct a satisfactory preflight/inspection. Individual techniques unsafe and/or in violation of established procedures.

2.3.10. **Area 10. Use of Checklist.**

2.3.10.1. Q. Consistently used correct checklist, gave correct responses and accomplished appropriate actions at the appropriate time throughout the mission.

2.3.10.2. Q-. Checklist responses were untimely and/or crew member required continual prompting for correct responses/action.

2.3.10.3. U. Used incorrect checklist or consistently omitted checklist items. Was unable to identify the correct checklist to use for a given situation. Omitted or did not complete checklist(s) at the appropriate time.

2.3.11. **Area 11. Forms/Reports/Logs.**

2.3.11.1. Q. All required pre/post flight forms and/or flight plans were complete, accurate, readable, and accomplished on time IAW applicable directives. Relayed an accurate debrief of significant events to applicable agencies (Intel, Weather, Maintenance, automated integrated radar control for air traffic (AIRCAT), etc.).

2.3.11.2. Q-. Minor errors on pre/post flight forms and/or flight plans did not affect conduct of the flight/mission. Incorrectly or incompletely reported some information due to minor errors, omissions, and/or deviations.

2.3.11.3. U. Did not accomplish required pre/post flight forms and/or flight plans. Omitted or incorrectly reported significant information due to major errors or omissions.

2.3.12. **Area 12. Personal/Professional Equipment/Flight Publications. Note:** Required flight publications are specified in AFI 11-2CV-22Vol 3, *CV-22 Operations Procedures*.

2.3.12.1. Q. Had all required personal/professional equipment. Displayed satisfactory knowledge of the care and use of such equipment and the contents of required publications. Required equipment inspections were current. Publications were current, contained all supplements/changes and were properly posted.

2.3.12.2. Q-. Did not have all required personal/professional equipment or had limited knowledge of the use or the content of required publications. Publications contained deficiencies that would not adversely impact flight safety or mission accomplishment.

2.3.12.3. U. Did not have required personal/professional equipment essential for the mission. Unsatisfactory knowledge of the care and use of equipment or the content of required publications. Equipment inspections were overdue or equipment was unserviceable. Publications were out dated and/or contained deficiencies that would adversely impact flight safety or mission accomplishment.

2.3.13. **Area 13. Emergency and Life Support Equipment/Procedures.**

2.3.13.1. Q. Satisfactory systems/procedural knowledge. Displayed satisfactory knowledge of location and use of emergency and life support equipment. Operated within prescribed limits and correctly diagnosed problems. Performed/explained proper wear, use, and corrective action for each type of equipment/malfunction. Effectively used available aids.

2.3.13.2. Q-. Marginal systems/procedural knowledge. Limited knowledge of location and use of emergency and life support equipment. Operated within prescribed limits but was slow to analyze problems or apply proper corrective actions. Did not effectively use, omitted, or deviated in use of checklist and/or available aids.

2.3.13.3. U. Unsatisfactory systems/procedural knowledge. Displayed unsatisfactory knowledge of emergency and life support equipment. Exceeded flight manual limitations. Unable or failed to analyze problem or take proper corrective action. Did not use checklist and/or available aids.

2.3.14. **Area 14. Briefings/Debriefings**

2.3.14.1. Q. Ensured briefing contained all applicable information. Prepared at briefing time. Briefings effectively organized and professionally presented in a logical sequence. Presented all objectives, training events and special interest items. Effectively used available briefing aids. Debriefed mission using specific, positive and non-threatening negative feedback of team and individual performance. Provided specific ways to correct errors. Asked for inputs from others. Re-capped key points and compared mission results with mission objectives. Properly filled out all appropriate forms.

2.3.14.2. Q-. Omitted items pertinent but not critical to the mission. Some difficulty communicating clearly. Did not make effective use of available briefing aids. Limited discussion of training events or special interest items. Dwelled on non-essential items. Not fully prepared for briefing. Debriefed mission without specific, positive and non-threatening negative feedback on individual and team performance. Did not consistently seek input from others. Incomplete or inadequate re-cap of key points and comparison of mission results to mission objectives.

2.3.14.3. U. Failed to conduct/attend required briefings. Failed to use appropriate briefing aids. Omitted essential items or did not correct erroneous information that could affect mission accomplishment. Demonstrated lack of knowledge of the subject. Briefing was poorly organized and not presented in a logical sequence. Presented erroneous information that would affect safe/effective mission accomplishment. Presentation created doubts or confusion. Failed to discuss training events or special interest items. Late to crew transport due to excessively long briefing. Did not provide positive and non-threatening negative feedback during debriefing. Did not seek input from others. Did not re-cap key mission points or compare mission results to mission objectives.

2.3.15. **Area 15. Communication.**

2.3.15.1. Q. Communicated using precise, standard terminology. Acknowledged all communications. Asked for/provided clarification when necessary. Stated opinions/ideas. Asked questions when uncertain. Advocated specific courses of action. Did not let rank affect mission safety.

2.3.15.2. Q-. Unclear or incomplete communication led to repetition or misunderstanding. Slow to ask for or give constructive feedback/clarifications. Inconsistent use of precise, standard terminology. Did not always state opinions/ideas, ask questions when uncertain, or make positive statements to flight members.

2.3.15.3. U. Failed to communicate effectively. Continuously interrupted others, mumbled, and/or conduct/attitude was detrimental to communication among crew members. Withheld information and failed to ask for/respond to constructive criticism. Failed to use precise, standard terminology. Repeatedly failed to acknowledge communications. Did not state opinions, ask questions when unsure or attempt to motivate flight members using positive statements.

2.3.16. **Area 16. Risk Management/Decision Making.**

2.3.16.1. Q. Identified contingencies and alternatives. Gathered and cross checked relevant data before deciding. Clearly stated problems and proposed solutions. Investigated doubts and concerns of crew members. Used facts to come up with solution. Involved and informed necessary crew members when appropriate. Coordinated mission crew activities to establish proper balance between command authority and crew member participation, and acted decisively when the situation required. Clearly stated decisions, received acknowledgement, and provided rationale for decisions.

2.3.16.2. Q-. Partially identified contingencies and alternatives. Made little effort to gather and cross check relevant data before deciding. Did not clearly state problems and propose solutions. Did not consistently use facts to come up with solutions. Did not effectively inform crew members when appropriate. Did not effectively coordinate mission crew activities to establish a proper balance between command authority and crew member participation, and acted indecisively at times.

2.3.16.3. U. Failed to identify contingencies and alternatives. Made no effort to gather and cross check relevant data before deciding. Did not inform necessary crew members when appropriate. Did not use facts to come up with solution. Avoided or delayed necessary decisions which jeopardized mission effectiveness. Did not coordinate mission crew activities to establish proper balance between command authority and crew member participation; acted indecisively.

2.3.17. **Area 17. Task Management.**

2.3.17.1. Q. Correctly prioritized tasks. Used available resources to manage workload. Asked for assistance when overloaded. Clearly stated problems and proposed solutions. Accepted better ideas when offered. Used facts to come up with solution. Clearly communicated and acknowledged workload and task distribution. Demonstrated high level of vigilance in both high and low workload conditions. Prepared for expected or contingency situations. Avoided the creation of self-imposed workload/stress. Recognized and reported work overloads in self and others.

2.3.17.2. Q-. Did not consistently or correctly prioritize tasks. Did not effectively use available resources to manage workload. Did not clearly communicate and acknowledge workload and task distribution. Did not consistently demonstrate high level of vigilance in both high and low workload conditions. Slow to prepare for expected or contingency situations. Created some self-imposed workload/stress due to lack of planning. Slow to recognize and report work overloads in self and others.

2.3.17.3. U. Failed to correctly prioritize tasks. Did not use available resources to manage workload. Did not communicate and acknowledge workload or task distribution. Did not demonstrate high level of vigilance in both high and low workload conditions. Extremely slow to prepare for expected or contingency situations. Created significant self-imposed workload/stress due to lack of planning. Failed to recognize and report work overloads in self and others.

Chapter 3

INSTRUCTOR EVALUATIONS

3.1. General. The instructor grading criteria apply to initial, requalification, and all periodic instructor evaluations. The examinee must demonstrate the ability to instruct in a safe and effective manner. **(T-2).**

3.2. Requirements. Evaluate instructors on areas listed in **Table 3.1** Instructor candidates must be qualified in all areas they will instruct and should have a solid understanding of systems, procedures, and techniques. **(T-2).** Initial instructor evaluations may be a stand-alone evaluation or accomplished in conjunction with a periodic qualification/mission evaluation. Accomplish periodic instructor evaluations in conjunction with periodic qualification/mission evaluations IAW AFMAN 11-202V2 (and applicable supplements). If able, evaluate instructor candidates instructing actual students. Otherwise, the flight examiner may act as the student. An instructor or evaluator graded Q-2 on any evaluation will not perform instructor or evaluator duties until additional training is completed. **(T-2).** Reference AFI 11-202 Vol 2 for restrictions to instructors who fail an evaluation for any item in **Table 3.1** The annotation RQ/INSTR will not be used for loss of qualification due to expiration of required periodic evaluation or loss of currency. **(T-3).** Only annotate RQ/INSTR if requalification is required due to a failure of instructor ability. For all other RQ evaluations, annotate as RQ (QUAL, MSN etc.)

3.2.1. Periodic. Qualified instructors will be evaluated on all areas per **Table 3.1** during all periodic evaluations. **(T-2).** Instructors will, at a minimum, demonstrate instructional ability and maneuvers/procedures in flight or in an ATD, all other areas may be evaluated via alternate method. **(T-2).**

3.2.2. There are no requisites for initial instructor evaluations. Requisites for periodic evaluations administered to qualified instructors will be completed IAW the requirements for the type (QUAL/INSTM/MSN) evaluation being administered. **(T-2).**

3.2.3. Initial Qualification. Accomplish the initial instructor evaluation on a mission that permits accomplishment of all required instructor areas. Initial instructor evaluations will be accomplished in the aircraft. **(T-2).**

3.3. Instrument. Instrument instructor evaluations may be accomplished in the aircraft or an ATD. Evaluate instructor pilot candidate's instructional ability during emergency and instrument procedures.

3.4. Qualification. Periodic and requalification evaluations may be accomplished in either the aircraft or an ATD per **paragraph 1.10** of this manual

3.5. Mission. Mission instructor evaluations are normally conducted in the aircraft, however periodic evaluations may be accomplished in an ATD based on operational necessity and with squadron commander approval annotated on the examinee's AF Form 8. Two consecutive mission evaluations will not be conducted in an ATD. **(T-3)**. Pilots must be aircraft commander qualified in a special mission prior to receiving instructor qualification/certification in that mission. **(T-2).**

Table 3.1. Instructor Evaluation Grading Areas (all Crew Positions).

Area	Notes	Grading Areas
20	2	Mission Preparation
21	1	Instructional Ability **(CRITICAL)**
22	2	Instructor Knowledge
23	2	Briefings/Debriefings/Critique/Forms
24	1	Demonstration of Maneuvers/Procedures
25-29		Reserved for future use
Notes: 1. Required in-flight or ATD 2. Required in-flight or alternate method		

3.6. Instructor Grading Criteria.

3.6.1. **Area 20. Mission Preparation.**

3.6.1.1. Q. Thoroughly reviewed student's training documentation. Ascertained student's present level of training. Assisted student in pre-mission planning and allowed student time for questions. Correctly prioritized training events. Gave student a clear idea of mission training objectives.

3.6.1.2. Q-. Did not thoroughly review student's training folder or correctly ascertain student's present level of training. Caused student to hurry pre-mission planning. Poorly prioritized training events. Training plan/scenario made poor use of time.

3.6.1.3. U. Did not review student's training folder. Did not ascertain student's present level of training. Did not assist student with pre-mission planning or did not allow time for questions. Did not prioritize training events. Failed to give student a clear idea of mission training objectives, methods, and sequence of events.

3.6.2. **Area 21. Instructional Ability. (CRITICAL)**

3.6.2.1. Q. Demonstrated proper instructor ability and communicated effectively. Provided appropriate guidance when necessary. Planned ahead, and provided accurate, effective, and timely instruction. Identified and corrected potentially unsafe maneuvers/situations.

3.6.2.2. U. Failed to effectively communicate or provide timely feedback. Performed or taught improper procedures/techniques/tactics to the student. Did not provide corrective action when necessary. Did not plan ahead or anticipate student problems. Did not identify unsafe maneuvers/situations in a timely manner. Made no attempt to instruct.

3.6.3. **Area 22. Instructor Knowledge.**

3.6.3.1. Q. Demonstrated a high level of knowledge of all applicable aircraft systems, techniques, procedures, missions, publications and tactics to be performed. Accurately completed appropriate training records. Comments were clear and pertinent.

3.6.3.2. Q-. Minor errors/deficiencies in knowledge of above areas did not affect safety or adversely affect student progress. Minor errors or omissions in training records. Comments were incomplete or slightly unclear.

3.6.3.3. U. Lack of knowledge of publications or procedures seriously detracted from instructor effectiveness. Could not apply knowledge of above areas. Did not complete required forms or records. Comments were invalid, unclear, or did not accurately document performance.

3.6.4. **Area 23. Briefings/Debriefings/Critique/Forms.**

3.6.4.1. Q. Briefings were well organized, accurate, and thorough. Reviewed student's present level of training and defined mission events to be performed. Showed a satisfactory ability during the critique to reconstruct the flight, offer mission analysis, and provide guidance where appropriate. Training grade reflected the actual performance of the student relative to the standard. Pre-briefed the student's next mission, if required. Properly completed training folders/records, understood grading policies and procedures.

3.6.4.2. Q-. Minor errors or omissions in briefings and/or critique did not affect safety or adversely affect student progress.

3.6.4.3. U. Briefings/debriefings were marginal or non-existent. Major errors or omissions in briefings/debriefings. Did not review student past performance. Analysis of events or maneuvers was incomplete, inaccurate, or confusing. Training grade did not reflect actual performance of student. Overlooked or omitted major discrepancies. Incomplete pre-briefing of student's next mission, if required. Improperly completed training folders/records, failed to understand grading policies and procedures.

3.6.5. **Area 24. Demonstration of Maneuvers/Procedures.**

3.6.5.1. Q. Effectively demonstrated procedures and techniques. Provided concise, meaningful, and timely in-flight commentary. Had thorough knowledge of applicable aircraft systems, procedures, publications, and instructions.

3.6.5.2. Q-. Performed required maneuvers/procedures with minor deviations from prescribed parameters. In flight commentary was sometimes unclear or poorly timed, interfering with student performance. Discrepancies in the above areas did not adversely affect safety or student progress.

3.6.5.3. U. Failed to properly perform required maneuvers/procedures. Made major procedural errors. Did not provide in flight commentary and/or in flight commentary was incorrect or unsafe. Insufficient knowledge of aircraft systems, procedures, and/or proper source material.

3.6.6. **Areas 25 - 29. Reserved for future use.**

Chapter 4

PILOT EVALUATIONS

4.1. General. All pilots require a QUAL and INSTM evaluation; these two evaluations are normally combined and can be completed in either the aircraft, an ATD or a combination of the two. **Note***:* The MV-22 is considered a different MDS, therefore, all pilots require a CV-22 QUAL/INSTM evaluation. NATOPS evaluation forms will be maintained in the pilot's FEF **(T-1).** MSN qualified pilots require a separate MSN evaluation. Evaluation standards will be administered in accordance with the individual's crew position as listed below: **(T-3). Note***:* The MV-22 is considered a different MDS, therefore, all pilots require a CV-22 QUAL/INSTM evaluation. NATOPS evaluation forms will be maintained in the pilot's FEF.

4.1.1. Basic Aircraft Qualified (BAQ) pilot (FP), Mission Pilot (MP):

4.1.1.1. FPs and MPs are evaluated to the standards outlined in **Table 2.1** and Tables **4.1, 4.2, 4.3**.

4.1.1.2. MPs certified as aircraft commanders should be evaluated as aircraft commanders and formation lead. This implies they have command of the aircraft, crew, and formation.

4.1.2. Evaluator Pilots (EP): EPs may be evaluated IAW **Table 4.1** and AFI 11-202V2.

4.2. Requirements. Refer to **Chapter 2** for general and **Chapter 3** for instructor grading areas and criteria. Pilot required areas and criteria follow in this chapter.

4.2.1. **Qualification (QUAL):**

4.2.1.1. See Tables **4.1** and **4.3** for required QUAL evaluation areas. Requisites (prerequisites for initial/re-qualification evaluations) include qualification open and closed book examinations (or formal school end of course examinations), EPE (requisite), and boldface examination. This evaluation can be accomplished in combination with the INSTM evaluation**.**

4.2.1.2. Initial QUAL evaluations will be accomplished in the aircraft. **(T-2).** Requalification evaluations should be conducted in the aircraft.

4.2.1.3. Periodic QUAL evaluations may be accomplished in the ATD.

4.2.1.4. The evaluation profile will include: visual flight rules (VFR) pattern; roll-on landing, normal and steep approach, airland, approach to hover and hovering maneuvers; engine-out go-around (simulated in aircraft), and simulated engine-out landing (simulated in aircraft). **(T-3).**

4.2.2. **Instrument (INSTM):**

4.2.2.1. See Tables **4.1** and **4.3** for required INSTM evaluation areas. The evaluation profile will include: One precision, one non-precision approach, holding or procedure turn, and a missed approach. **(T-3)**. Both Initial (INIT) and recurring INSTM evaluations may be accomplished in either the aircraft or an ATD.

4.2.2.2. The instrument written examination is a prerequisite for initial and requalification evaluations. It is a requisite for periodic evaluations.

4.2.2.3. The instrument and qualification flight evaluations may be combined.

4.2.3. **Mission (MSN):**

4.2.3.1. See Tables **4.2** and **4.3** for MSN evaluation areas and subparagraphs below for requirements. Requisites (prerequisites for initial/re-qualification evaluations) include Mission Open and Closed Book examinations (or formal school end of course examinations) and EPE.

4.2.3.2. Evaluations will be accomplished at night. **(T-2).**

4.2.3.3. For all mission evaluations, flight examiners will ensure evaluation profiles include demonstration of an adequate number of events to thoroughly measure knowledge of specific employment procedures to include tactical defensive measures. **(T-3).** One form of low level flight will be accomplished every other year. **(T-3).**

4.2.4. **Special Mission Event (SME):**

4.2.4.1. SMEs are listed in AFI 11-2CV-22V1, *CV-22 Aircrew Training*.

4.2.4.2. See **Table 4.2** for SME evaluations.

4.2.4.3. Initial SME evaluations may be conducted separately or in conjunction with a mission evaluation.

4.2.4.4. A SME current and qualified pilot that accomplishes a SME during a MSN evaluation can be credited a SME evaluation completion. The MSN and SME evaluations are annotated separately on the AF Form 8.

4.2.4.5. There are no requisites for SME evaluations.

4.2.4.6. Annotate the specifics of SME evaluations in the Block IV, Comments section of the AF FORM 8/8a.

Table 4.1. Pilot INSTM/QUAL Grading Areas.

Area	Notes	Grading Areas	QUAL	INSTM
30	1	Ground Operations/Taxi	X	
31	1,5	Takeoff 60 or 75 Short Take Off (STO) or Hover	X	
32	1	Instrument Departure		X
33	1	En route Navigation/Use of Navigational Aids (NAVAID)		X
34	1	Descent/Arrival Procedures		X
35	1	Holding/Procedure Turn		X
36	1,2,3	Precision Approach (PAR or Instrument Landing System (ILS))		
36a	1	PAR		X
36b	1	ILS		X
37	2,3	Non-Precision Approach (TACAN, VOR, LOC, Airport Surveillance Radar (ASR))		
37a	1	Tactical Air Navigation (TACAN)		X
37b	1	VOR		X

Area	Notes	Grading Areas	QUAL	INSTM
37c	1	LOC		X
37d	1	ASR		X
38	1	Circling/Side-Step Approach		X
39	1	Engine-Out Approach		X
40	1	Missed Approach/Go-Around		X
41	1	Engine-Out Go-Around	X	
42	1	VFR Pattern	X	
43	1	Final Approach and Landing	X	
43a	1	Roll on Landing	X	
43b	1	Airland	X	
43c	1	Normal Approach	X	
43d	1	Steep Approach	X	
43e	1	Hovering Maneuvers	X	
43f	1	Single Engine Landing	X	
44		Systems Operations/Knowledge/Limitations/NAS	X	X
45-49		Reserved for future use		

Notes:

1. Required in-flight or simulator.
2. Only one of the two required approaches may be controller directed (PAR/ASR).
3. Any one required.
4. One STO is required.

Table 4.2. Pilot MSN and SME Grading Areas.

Area	Notes	Grading Areas	MSN	SME
50	1	Low Level Operations/Navigation		
50a	1, 2	Low Level – Terrain Following (TF)	X	
50b	1, 2	Low Level – Night Vision Goggle (NVG)	X	
51	2	Threat Avoidance/Tactics/Countermeasures	X	
52	3	Formation	X	
53	1	NVG Tactical Approach and Landing	X	
54	2	Air Refueling	X	
55	1, 4	Low Visibility Approach (LVA)	X	
56	1	Time On Target	X	
57	1	Alternate Insertion/Extraction	X	
58	3	Systems Operations/Knowledge/Limitations	X	
59	3	Procedures (Norm/Emerg)	X	
60-99		Reserved for future use		

Area	Notes	Grading Areas	MSN	SME
Notes: 1. Required in-flight. (Every other evaluation for Area 50). 2. Required in-flight or simulator certified for this event. 3. Required in-flight or alternate method. 4. May be sampled via water operations, actual brown-out/white-out conditions, or unaided (NVGs up, Forward Looking Infrared (FLIR) off) approach to a blacked-out runway. Unaided approach to a blacked-out runway should not be accomplished on consecutive evaluations.				

Table 4.3. General Criteria.

Q	Altitude	+/- 200 feet
	Airspeed	+/- 10 knots
	Course	+/- 5 degrees/3 NM (whichever is greater)
	Arc	+/- 2 NM
Q-	Altitude	+/- 300 feet
	Airspeed	+/- 15 knots
	Course	+/- 10 degrees/5 NM (whichever is greater)
	Arc	+/- 3 NM
U		Exceeded Q- limits

4.3. Grading Criteria. The following subparagraphs contain grading criteria for the areas listed in **Tables 4.1** and **Table 4.2** The general criteria in **Table 4.3** apply during all phases of flight except as noted for specific events and instrument final approaches.

4.3.1. **Area 30. Ground Operations/Taxi.**

4.3.1.1. Q. Established and adhered to engine start, taxi, and take-off time to assure thorough preflight, check of personal equipment, crew/passenger briefings, etc. Accurately determined readiness of aircraft for flight. Completed all systems preflight/post flight inspections, and checklists IAW flight manual. Conducted taxi operations according to flight manual, AFI 11-218, *Aircraft Operations and Movement on the Ground*, and local procedures.

4.3.1.2. Q-. Same as above except for minor procedural deviations that did not detract from mission effectiveness.

4.3.1.3. U. Failed to accurately determine readiness of aircraft for flight. Major deviations in procedure that would preclude safe mission accomplishment. Errors directly contributed to a late takeoff that degraded the mission or made it ineffective. Omitted checklist items.

4.3.2. **Area 31. Takeoff.**

4.3.2.1. Q. Maintained smooth, positive aircraft control throughout takeoff. Performed takeoff in accordance with flight manual and as published/directed.

4.3.2.2. Q-. Minor deviations from published procedures without affecting safety of flight. Aircraft control was safe but not consistently smooth and positive. Hesitant in application of procedures or corrections.

4.3.2.3. U. Takeoff was potentially dangerous. Exceeded aircraft/systems limitations. Failed to establish proper climb attitude. Excessive deviation from intended flight path. Violated flight manual procedures. Exceeded Q- criteria.

4.3.3. **Area 32. Instrument Departure.**

4.3.3.1. Q. Performed departure IAW published procedures and directives. Complied with all restrictions or controlling agency instructions. Made all required reports. Applied course/heading corrections promptly. Demonstrated smooth, positive aircraft control.

4.3.3.2. Q-. Minor deviations in navigation occurred during departure. Slow to comply with controlling agency instructions or unsure of reporting requirements. Slow to apply course/heading corrections. Aircraft control was not consistently smooth and positive.

4.3.3.3. U. Instrument departure was not in accordance with technical orders, directives, or published procedures. Failed to comply with published/directed departure, or controlling agency instructions. Accepted an inaccurate clearance. Aircraft control was erratic.

4.3.4. **Area 33. En Route Navigation/Use of NAVAIDs.**

4.3.4.1. Q. Able to navigate using all available means. Used appropriate navigation procedures. Ensured navaids were properly tuned, identified, and monitored. Complied with clearance instructions. Aware of position at all times. Remained within the confines of assigned airspace.

4.3.4.2. Q-. Minor errors in procedures/use of navigation equipment. Some deviations in tuning, identifying, and monitoring navaids were observed. Slow to comply with clearance instructions. Had some difficulty in establishing exact position and course. Slow to adjust for deviations in time and course.

4.3.4.3. U. Major errors in procedures/use of navigation equipment. Did not ensure navaids were tuned, identified and monitored. Could not establish position. Failed to recognize checkpoints or adjust for deviations in time and course. Did not remain within the confines of assigned airspace. Exceeded Q- criteria.

4.3.5. **Area 34. Descent/Arrival Procedures.**

4.3.5.1. Q. Performed descent as directed. Complied with all flight manual, controller issued, or standard terminal arrival route restrictions in a proficient manner. Accomplished all required checks.

4.3.5.2. Q-. Performed descent as directed with minor deviations that did not compromise mission safety. Slow to comply with controller instructions and accomplish required checks.

4.3.5.3. U. Performed descent with major deviations. Failed to follow controller instructions or made erratic corrections. Exceeded flight manual limitations or did not accomplish required checks.

4.3.6. **Area 35. Holding/Procedure Turn.**

4.3.6.1. Q. Performed entry and holding in accordance with published procedures and directives. Holding pattern limits exceeded by not more than:

4.3.6.1.1. Very High Frequency Omni Directional Range Station (VOR) Leg timing: +/- 15 seconds

4.3.6.1.2. TACAN: +/- 2 NM

4.3.6.2. Q-. Performed entry and holding procedures with minor deviations. Holding pattern limit exceeded by not more than:

4.3.6.2.1. VOR Leg timing: +/- 30 seconds.

4.3.6.2.2. TACAN: +/- 3 NM.

4.3.6.3. U. Holding was not in accordance with technical orders, directives, or published procedures. Exceeded Q- holding pattern limits.

4.3.7. **Area 36. Precision Approach (PAR or ILS). Note:** Use the following criteria as general tolerances for airspeed, altitude, heading, glide slope, and azimuth. Airspeed tolerances are based on computed approach speed.

4.3.7.1. Q.

4.3.7.1.1. Heading: +/- 5 degrees of controller's instructions (PAR)

4.3.7.1.2. Glide slope: Within one dot (ILS)

4.3.7.1.3. Azimuth: Within one dot (ILS)

4.3.7.2. Q-.

4.3.7.2.1. Heading: +/- 10 degrees of controller's instructions (PAR)

4.3.7.2.2. Glide slope: Within one dot low, two dots high (ILS), after runway was in sight. Examinee momentarily deviated below glide path but corrected for a safe landing.

4.3.7.2.3. Azimuth: Within two dots (ILS)

4.3.7.3. U.

4.3.7.3.1. Exceeded Q- criteria.

4.3.7.4. **Subarea 36a. Precision Approach Radar.**

4.3.7.4.1. Q. Approach was IAW flight manual, directives and published procedures. Smooth and timely response to controller's instructions. Established initial glide path and maintained glide slope with minor deviations. Complied with decision height. Position would have permitted a safe landing. Elevation did not exceed slightly above or slightly below glide path.

4.3.7.4.2. Q-. Performed approach with minor deviations. Slow to respond to controller's instructions and make corrections. Position would have permitted a safe landing. Elevation did not exceed well above or well below glide path.

4.3.7.4.3. U. Approach not IAW flight manual, directives or published procedures. Erratic course and glide slope corrections. Did not make corrections or react to controller's instructions. Did not comply with decision height and/or position would not have permitted a safe landing. Exceeded Q- limits.

4.3.7.5. **Subarea 36b. Instrument Landing System.**

4.3.7.5.1. Q. Approach was IAW flight manual, directives, and published procedures. Smooth and timely corrections to azimuth and glide slope. Complied with decision height and position permitted a safe landing.

4.3.7.5.2. Q-. Performed procedures with minor deviations. Slow to make corrections or initiate procedures. Slow to comply with decision height. Position would have permitted a safe landing.

4.3.7.5.3. U. Approach not IAW flight manual, directives, or published procedures. Erratic course/glide slope corrections. Did not comply with decision height or position would not have permitted a safe landing. Exceeded Q- criteria.

4.3.8. **Area 37. Non-Precision Approach (TACAN, VOR, LOC, ASR). Note*:* Use the following criteria for Areas 37a-37d.**

4.3.8.1. Q. Approach was IAW flight manual, directives, and published procedures. Used appropriate descent rate to arrive at Minimum Decision Altitude (MDA) at or before Visual Descent Point (VDP) (if applicable). Position permitted a safe landing. Smooth and timely response to controller's instructions (ASR).

4.3.8.1.1. Heading: +/- 5 degrees (ASR).

4.3.8.1.2. Course: +/- 5 degrees at Missed Approach Point (MAP) (TACAN, VOR), less than one dot deflection (LOC).

4.3.8.1.3. MDA: +100/-0 feet.

4.3.8.1.4. MAP: Timing computed/adjusted within 10 seconds or distance within +/- .5 NM.

4.3.8.2. Q-. Performed approach with minor deviations. Arrived at MDA at or before the MAP, but past the VDP (if applicable). Position would have permitted a safe landing. Slow to respond to controller's instructions and make corrections (ASR).

4.3.8.2.1. Heading: +/- 10 degrees (ASR).

4.3.8.2.2. Course: +/- 10 degrees at MAP (TACAN, VOR).

4.3.8.2.3. Localizer: Within two dots deflection.

4.3.8.2.4. MDA: +150/-50 feet.

4.3.8.2.5. MAP: Timing computed/adjusted within 20 seconds or distance within + 1/ - .5 NM.

4.3.8.3. U. Approach not IAW flight manual, directives, or published procedures. Maintained steady-state flight below the MDA, and exceeded the -50 foot limit. Could not land safely from approach and did not initiate missed approach/go-around when appropriate or directed. Exceeded Q- criteria.

4.3.9. Area 38. **Circling/Side-Step Approach.**

4.3.9.1. Q. Remained within the lateral limits of circling category. Complied with controller's instructions. Attained runway alignment without excessive bank angles. Did not descend from the MDA until in a position to place the aircraft on a normal glide path or execute a normal landing.

4.3.9.1.1. Altitude: +100/-0 feet.

4.3.9.2. Q-. Slow to comply with controller's instructions. Attained runway alignment but occasionally required excessive bank angles or maneuvering.

4.3.9.2.1. Altitude: +150/-50 feet.

4.3.9.3. U. Exceeded the lateral limits of circling airspace. Did not comply with controller's instructions. Excessive maneuvering to attain runway alignment was potentially unsafe. Descended from the MDA before the aircraft was in position for a normal glide path or landing. Exceeded Q- criteria.

4.3.10. **Area 39. Engine-Out Approach. Note*:*** Use approach criteria for the type of approach being flown and the following.

4.3.10.1. Q. Performed procedures IAW the flight manual and associated directives. Individual technique complied with established procedures.

4.3.10.2. Q-. Unnecessary maneuvering due to minor errors in planning or judgment.

4.3.10.3. U. Major/unsafe deviations from procedures. Individual technique unsafe or violated established procedures. Potentially unsafe maneuvering.

4.3.11. **Area 40. Missed Approach/Go-Around.**

4.3.11.1. Q. Executed missed approach IAW published procedures and restrictions. Initiated and performed go-around promptly. Complied with controller's instructions. Applied smooth control inputs. Attained and maintained a positive climb.

4.3.11.2. Q-. Executed missed approach with minor deviations to published procedures/ directives. Was slow or hesitant to initiate go-around. Slow to respond to controller's instructions. Slightly over-controlled the aircraft. Allowed aircraft to descend during transition.

4.3.11.3. U. Did not execute missed approach IAW technical orders, directives or published procedures. Did not comply with controller's instructions. Deviations or misapplication of procedures could have led to an unsafe condition including excessive descent during transition. Exceeded Q- criteria.

4.3.12. **Area 41. Engine-Out Go-Around. Note*:*** Use Area 40 criteria and the following.

4.3.12.1. Q. Applied smooth control inputs.

4.3.12.2. Q-. Individual techniques were safe, but detracted from the maneuver.

4.3.12.3. U. Individual technique unsafe or violated established procedures.

4.3.13. **Area 42. VFR Pattern.**

4.3.13.1. Q. Adhered to published restrictions/local guidance. Performed traffic pattern and turn to final/final approach IAW flight manual procedures. Aircraft control was smooth and positive. Very slight over/under-shoot of final approach. Constantly cleared area of intended flight.

4.3.13.2. Q-. Minor deviations from published restrictions/local guidance. Performed traffic pattern and turn to final/final approach with minor deviations to procedures. Aircraft control was safe but not consistently smooth and positive. Over/under-shot final approach but was able to intercept a normal glide path. Adequately cleared area of intended flight.

4.3.13.3. U. Major/unsafe deviations from published restrictions/local guidance. Did not perform traffic pattern and turn to final/final approach IAW technical orders, directives or published procedures. Displayed erratic aircraft control. Over/under-shot final approach by a wide margin requiring a go-around or potentially unsafe maneuvering on final. Did not clear area of intended flight. Exceeded Q- criteria.

4.3.14. **Area 43. Final Approach and Landing. For Areas 43a through 43f.** Use the following criteria. **Note:** For instrument approaches, the examinee should utilize a normal glide slope from either the decision height or from a point where visual acquisition of the runway environment/intended point of landing is made.

4.3.14.1. Q. Performed the approaches and landings IAW the procedures and limitations outlined in the flight manual and published directives. Aircraft control was smooth and positive. Started the final descent on the desired approach angle. Demonstrated satisfactory control to maintain/correct to the desired rate of descent and approach angle. Touchdown/termination was within the desired area.

4.3.14.2. Q-. Same as Q except for minor deviations to procedures and limitations outlined in the flight manual and published directives.

4.3.14.3. U. Major deviations to the procedures and limitations outlined in the flight manual and published directives. Aircraft control was erratic/unsafe. Consistently overshot/undershot final approach. Failed to recognize/maintain the correct or desired rate of descent and approach angle. Failed to touchdown or terminate within the desired/briefed area. Exceeded the limits of flight manual.

4.3.15. **Area 44. Systems Operation/Knowledge/Limitations/National Airspace System (NAS).**

4.3.15.1. Q. Demonstrated/explained a complete knowledge of aircraft systems operations/ limitations and proper procedural use of systems. Demonstrated adequate knowledge of and complied with NAS rules and procedures in all areas of mission planning and flight operations.

4.3.15.2. Q-. Marginal knowledge of aircraft systems operations and limitations in some areas. Used individual technique instead of established procedures and was unaware of differences. Marginal knowledge of NAS rules and procedures.

4.3.15.3. U. Unsatisfactory systems knowledge. Failed to demonstrate/explain the procedures for aircraft system operations. Unsatisfactory knowledge of NAS rules and procedures.

4.3.16. **Areas 45 - 49. Reserved for future use.**

4.3.17. **Area 50. Low Level Operations/Navigation.**

4.3.17.1. **Area 50a. Low Level – TF.**

4.3.17.2. Q. Planned and flew a route to minimize risk to aircraft and crew for a given mission using terrain following procedures IAW flight manual, governing directives and appropriate Tactics, Techniques, and Procedures (TTP). Maintained altitude profile with minor deviations. Planned low altitude warnings prefaced with announcement to crew. Timely and appropriate response to unplanned low altitude warnings or obstacle warnings. Maintained situational awareness as to route location, distance from planned route of flight, minimum safe altitudes (MSAs), while effectively navigating aircraft to objective. Navigated properly to target.

4.3.17.3. Q-. Planned and flew a route to minimize risk to aircraft and crew for a given mission with minor deviations from terrain following procedures IAW flight manual, governing directives and appropriate TTP. Maintained altitude profile with multiple unannounced deviations. Planned low altitude warnings usually prefaced with announcement to crew. Timely and appropriate response to unplanned low altitude warnings or obstacle warnings. Momentary losses in situational awareness as to route location, poor navigation, distance from planned route of flight, minimum safe altitudes (MSAs), while effectively navigating aircraft to objective.

4.3.17.4. U. Excessive amount and/or excessive low altitude warnings. Major/unsafe deviations from established directives and appropriate TTP. Failed to maintain situational awareness as to route location, distance from planned route of flight, minimum safe altitudes (MSAs) an overall navigational situation. Failed to navigate properly to target.

4.3.18. **Area 50b. Low Level – NVG.**

4.3.18.1. Q. Could satisfactorily determine position using navigation systems and maps. Recognized all check/turn points. Remained within MSA corridor unless deviating for obstacles/threats, and stayed within range/area boundaries. Effectively used terrain masking to degrade/avoid threats. Demonstrated the capability to quickly adjust for deviations in timing and course.

4.3.18.2. Q-. Slow to determine the position using navigation systems or map. Slow to recognize check/turn points. Remained within MSA corridor unless deviating for obstacles/threats, and stayed within range/area boundaries. Satisfactorily used terrain masking to degrade/avoid threats. Slow to recognize and adjust for deviations in timing and course corrections.

4.3.18.3. U. Could not establish position. Failed to recognize check/turn points or adjust for deviations in timing and course. Exceeded the parameters for Q-. Flew outside the established range/area boundaries. Failed to use terrain masking.

4.3.19. **Area 51. Threat Avoidance/Tactics/Countermeasures.**

4.3.19.1. Q. Used proper terrain masking to limit aircraft susceptibility or highlighting to threats. Threat reactions were timely and correct. Accomplished appropriate countermeasures and performed maneuvers to counter the threat. Thoroughly familiar with threat indications visually and on defensive countermeasure equipment.

4.3.19.2. Q-. Limited use of terrain masking. Threat reactions were slow or inconsistent. Slow to accomplish appropriate countermeasures or perform maneuvers to counter threat. Limited knowledge of threat indications visually and on defensive countermeasure equipment.

4.3.19.3. U. Failed to use terrain masking or take the appropriate evasive action and/or activate the appropriate countermeasures. Exceeded the aircraft limitations during evasive maneuvering. Improper threat call and/or clearing/scanning procedures. Flew back through the threat area after completion of the evasive maneuver.

4.3.20. **Area 52. Formation.**

4.3.20.1. Q. Established appropriate formations. Effectively directed the flight to accomplish mission objectives. Good situational awareness and wingman consideration. Positive control of the flight/element. Smooth on the controls. Planned ahead and made timely decisions. Complied with established procedures. Maintained position with only momentary deviations. Made smooth and immediate position corrections. Maintained safe separation and complied with established procedures. Smooth, timely join-up. Good situational awareness.

4.3.20.2. Q-. Adequate flight management. Fair situational awareness and wingman consideration. Control inputs were not unsafe, but made it difficult for wingman to maintain position. Did not always plan ahead and/or hesitant in making decisions. Minor deviations in established procedures. Varied position, but within limits. Minor over-controlling. Minor procedural deviations. Slow join-up. Fair situational awareness.

4.3.20.3. U. Did not establish the appropriate formations. Poor situational awareness and wingman consideration. Rough on the controls. Major deviations in established procedures. Indecisive. Unable to maintain a formation position. Abrupt position corrections. Significant over-controlling requiring assistance from other pilot. Did not maintain safe separation. Unsafe join-up or formation procedures. Poor situational awareness.

4.3.21. **Area 53. NVG Tactical Approach and Landing.**

4.3.21.1. Q. Properly briefed the crew about approach intentions. Maintained controlled, stable approach without excessive deviations. Able to perform an NVG landing or Alternate Insertion/Extraction (AIE) to the intended point of landing. Able to apply techniques and procedures from AFTTP 3-3 *CV-22 Combat Aircraft Fundamentals* and adjust parameters based on external factors such as terrain and wind.

4.3.21.2. Q-. Briefed the crew about approach intentions with minor omissions. Maintained controlled, stable approach with minor deviations to the intended point of landing. Able to apply techniques and procedures from AFTTP 3-3 *CV-22 Combat Aircraft Fundamentals*, but had difficulty adjusting parameters based on external factors such as terrain and wind.

4.3.21.3. U. Exceeded standards for Q-.

4.3.22. **Area 54. Air Refueling.**

4.3.22.1. Q. Performance IAW the procedures outlined in the flight manual and other published directives. Aircraft control was smooth and positive during the rendezvous/join-

up/contact/disconnect/crossover. Contacts were accomplished in a timely fashion, with controlled misses (if applicable). When not flying, closely monitored fuel management, aircraft systems/instruments and lighting. Assisted the pilot flying as briefed/required.

4.3.22.2. Q-. Same as Q above except for minor deviations which did not affect safety/mission accomplishment.

4.3.22.3. U. Major deviations to the procedures outlined in the flight manual and other published directives. Aircraft control was erratic/unsafe. Unable to perform contacts and/or misses were dangerous. When not flying, failed to monitor fuel management, aircraft systems/instruments and/or aircraft flight path/position. Did not assist the pilot flying as briefed/required.

4.3.23. **Area 55. Low Visibility Approach**

4.3.23.1. Q. Without outside visual references, performed smooth transition from stabilized final approach altitude and ground speed to pre-briefed location +/- .02 NM and parameters. Ensured a controlled descent rate with minimal drift at touchdown. Maintained stable platform at briefed altitude and ground speed if applicable. Made appropriate corrections to heading and drift. Called for or accomplished the appropriate checklists.

4.3.23.2. Q-. Without outside visual references, performed somewhat rough but effective transition from stabilized final approach altitude and ground speed to pre- briefed location +/- .05NM and parameters. Erratic descent rate to the ground with erratic but controlled drift. Maintained mostly stable platform at briefed altitude and ground speed if applicable. Some delay in appropriate corrections to heading and drift. Called for or accomplished the appropriate checklists.

4.3.23.3. U. Exceeded standards for Q-

4.3.24. **Area 56. Time On Target (TOT).**

4.3.24.1. Q. Maneuvered aircraft/formation appropriately based on outside factors in order to arrive at objective on-time +/- 30 seconds.

4.3.24.2. Q-. Maneuvered aircraft/formation with minor deviations in sound tactical flight in order to arrive at objective on-time +/- 60 seconds.

4.3.24.3. U. Failed to appropriately maneuver aircraft/formation. Failed to arrive at objective +/- 60 Seconds.

4.3.25. **Area 57. Alternate Insertion/Extraction.**

4.3.25.1. Q. Properly briefed the crew about AIE intentions. Established and maintained hover +/- 10 feet over desired location with tail scanner input, and +/- 10 degrees heading. Acknowledged and responded to crew input for changes to location or altitude. Responded to emergencies appropriately.

4.3.25.2. Q-. Briefed the crew about AIE intentions with minor omissions. Able to maintain hover +/- 15 feet over desired location with tail scanner input, and +/- 15 degrees heading. Slow to acknowledge and/or respond to crew input for changes to location or altitude. Slow but safe response to emergencies.

4.3.25.3. U. Exceeded Q- tolerances. Failed to acknowledge and respond to crew input for changes to location or altitude. Failed to respond to emergencies appropriately.

4.3.26. **Area 58. Systems Operation/Knowledge/Limitations.**

4.3.26.1. Q. Demonstrated/explained a complete knowledge of aircraft systems operations/limitations and proper procedural use of systems.

4.3.26.2. Q-. Marginal knowledge of aircraft systems operations and limitations in some areas. Used individual technique instead of established procedures and was unaware of differences.

4.3.26.3. U. Unsatisfactory systems knowledge. Failed to demonstrate/explain the procedures for aircraft system operations.

4.3.27. **Area 59. Procedures (Norm/Emerg)**

4.3.27.1. Q. Performed/demonstrated/explained both normal and emergency flight and/or ground procedures IAW applicable directives.

4.3.27.2. Q-. Marginal performance/demonstration/explanation of flight and our ground procedures or was unaware of procedures or where procedures are located and explained.

4.3.27.3. U. Failed to properly perform/demonstrate/explain appropriate procedures.

4.3.28. **Areas 60 - 99. Reserved for future use.**

Chapter 5

FLIGHT ENGINEER (FE) EVALUATIONS

5.1. General. Table 5.1 contains flight engineer QUAL/MSN/SME grading areas. BAQ only first flight engineers (FF) require a QUAL evaluation. Mission qualified flight engineers (MF) will accomplish a combined Qual/MSN evaluation. **(T-2).** Instructors will meet the requirements of Chapter 3 and demonstrate instructor ability on all periodic evaluations as outlined in Table 3.1. **(T-2).**

5.2. Requirements. Evaluate all general areas outlined in Table 2.1 on all evaluations. Also, evaluate all instructors on instructor areas in Table 3.1 FE specific areas and evaluation criteria are listed in Table 5.1**;** areas marked by an X are required in order for the specific evaluation to be completed.

5.3. QUAL Evaluations (QUAL). This type of evaluation is for FEs that maintain BAQ status only. See Table 5.1 for required QUAL evaluation requirements. QUAL evaluations may be accomplished during daylight hours.

5.4. Combined Qualification/Mission Evaluations (QUAL/MSN). QUAL/MSN evaluations will be accomplished at night. **(T-2).** For all QUAL/MSN evaluations, flight examiners will ensure evaluation profiles include demonstration of adequate events to thoroughly measure knowledge of specific employment procedures. **(T-3).** BAQ flight engineers upgrading to mission flight engineers will complete a full QUAL/MSN evaluation to realign their periodic expiration date as appropriate. **(T-2).**

5.4.1. Hoist/Alternate Insertion and Extraction (AIEs).

5.4.1.1. Hoist/AIE requirements. Initial and periodic evaluations require both a hoist and a fast rope be accomplished.

5.4.1.2. For periodic evaluations, if the evaluate is certified, live AIEs are encouraged, but not required.

5.4.2. Special Mission Event(SME): SMEs are listed in AFI 11-2CV-22Vol 1. Initial SME evaluations may be conducted separately or in conjunction with a mission evaluation. If a FE is current and qualified and the SME is accomplished during the evaluation profile, the SME is evaluated during the MSN evaluation but annotated separately on the AF Form 8/8a.

5.4.2.1. Ramp Mounted Weapon System. Once trained and evaluated on one gun system during the initial SME evaluation, subsequent certification on any other system does not require an evaluation. Document certification IAW AFI 11-2CV-22 Vol 1.

5.4.2.2. FEs qualified in more than one gun system may be evaluated on either system. Verbally evaluate the system not evaluated by performance. Do not verbally evaluate the same system on subsequent evaluations.

Table 5.1. Flight Engineer QUAL/MSN/SME Grading Areas.

Area	Notes	Grading Areas	QUAL	MSN	SME
200	1	Weight and Balance/Takeoff and Landing Data (W&B/TOLD)	X		

Area	Notes	Grading Areas	QUAL	MSN	SME
201	3	Before Exterior/Exterior/Interior	X		
202	2,3	Before Starting Engines/Starting Engines	X		
203	2	Before Taxi/Taxi	X		
204	2	Before Takeoff	X		
205	2	After Takeoff	X		
206	2,3	Climb/Descend			
207	2	Before Landing	X		
208	3	After Landing	X		
209	2,3	Engine Shutdown	X		
210	1	Cabin Configuration/Loading	X		
211	1	Cryptological System Operations		X	
212	3	Low Level Navigation			
213		Formation Procedures		X	
214	3	TF/Terrain Avoidance (TA) Procedures			
215		Degraded Systems			
216		Infil/Exfil Procedures		X	
217	1,5	Alternate Insertion/Extraction Procedures		X	
218	3	Air Refueling			
219	4	Ramp Mounted Weapon System (or alternate Weapon System)		X	X
220	2	Low Visibility Approach		X	
221	3	Systems Operations/Knowledge/Limitations	X	X	
222	3	Aircraft Defensive Systems		X	
223	1	Night Water Hoist			X
224 - 249		Reserved for future use			

Notes:

1. Required in-flight.

2. Required in-flight or simulator certified for this event.

3. Required in-flight or alternate method. Refer to Paragraph 1.10.2.

4. Alternate between weapon system positions (every other evaluation). Alternate method should be used on weapon system not fired.

5.5. Grading Criteria. The following subparagraphs contain grading criteria for the areas listed in **Table 5.1**.

5.5.1. **Area 200.** Weight and Balance/Take-off and Landing Data (W&B/TOLD).

5.5.1.1. Q. Correctly computed the W&B/TOLD data using applicable performance charts and corrections for existing field conditions. Was fully knowledgeable of W&B/TOLD calculations.

5.5.1.1.1. TOLD criteria:

5.5.1.1.1.1. Required Short Takeoff and Landing (STOL) Airspeeds: +/- 2 knots.

5.5.1.1.1.2. Required STOL Distances: +/- 200 feet.

5.5.1.1.1.3. Hover Mast Torque Required: +/- 2%

5.5.1.1.1.4. Single Engine Service Ceiling Data: +/- 500 feet.

5.5.1.1.2. W&B criteria:

5.5.1.1.2.1. Center of Gravity +/- 0.5 inches

5.5.1.1.2.2. Takeoff or Landing Gross Weights +/- 500 lbs.

5.5.1.2. Q-. Minor errors in the appropriate use of W&B publications and performance charts resulting in data exceeding Q criteria. Had some knowledge of W&B/TOLD calculations. Would not have compromised safety of flight.

5.5.1.2.1. **TOLD criteria:**

5.5.1.2.1.1. Required STOL Airspeeds: +/- 4 knots.

5.5.1.2.1.2. Required STOL Distances: +/- 400 feet.

5.5.1.2.1.3. Hover Mast Torque required: +/- 4%.

5.5.1.2.1.4. Single Engine Service Ceiling Data: +/- 1,000 feet.

5.5.1.2.2. **W&B criteria:**

5.5.1.2.2.1. Center of Gravity: +/- 0.7 inches.

5.5.1.2.2.2. Takeoff or Landing Gross Weights; +/- 1,000.

5.5.1.3. U. Failed to compute W&B/TOLD data. Errors caused W&B/TOLD calculations to exceed Q- criteria. Limited knowledge of W&B/TOLD calculations. Did or could have compromised safety of flight.

5.5.2. **Areas 201 through 209. Reference Table 5.1** Use the following criteria.

5.5.2.1. Q. Accomplished required checklists without errors, omissions, or deviations and with accurate and timely responses. Backed up pilots on flight parameters (i.e., altitudes, airspeeds, and clearances). Recognized and corrected minor omissions or deviations. Recognized, reported, and properly documented out of limit conditions and aircraft malfunctions. Familiar with checklists and contents.

5.5.2.2. Q-. Accomplished required checklists with minor errors, omissions, or deviations and/or was slow to respond. Backed up pilots on flight parameters (i.e., altitudes, airspeeds, and clearances) with some deviations. Slow to recognize and correct minor omissions or deviations. Slow to recognize, report and properly document out of limit conditions and aircraft malfunctions.

5.5.2.3. U. Failed to use the proper checklist or made numerous errors, omissions or deviations. Failed to back up pilots on flight parameters (i.e., altitudes, airspeeds, and clearances). Allowed limitations to be exceeded and failed to recognize and report aircraft malfunctions.

5.5.3. **Area 210. Cabin Configuration/Loading.**

5.5.3.1. Q. Ensured the cabin was properly configured to accommodate mission requirements. Familiar with various configurations as outlined in applicable directives and properly stowed configuration items that were not used. Made sure all cargo items were loaded properly and secured in accordance with the cargo loading manual. Correctly briefed and performed passenger handling duties.

5.5.3.2. Q-. Difficulty configuring the aircraft but did not impede mission. Limited knowledge of various configurations as outlined in applicable directives, and stowed unused items with minor errors. Made sure all cargo items were loaded properly and secured in accordance with the cargo loading manual with minor errors and deviations. Correctly briefed and performed passenger handling duties with minor errors and deviations.

5.5.3.3. U. Failed to ensure proper aircraft configuration or caused mission delays. Had unsatisfactory knowledge of configurations and failed to properly stow configuration items. Incorrectly loaded and secured cargo items. Failed to brief and/or did not perform proper passenger handling procedures.

5.5.4. **Area 211. Cryptological System Operations.**

5.5.4.1. Q. Thorough knowledge of applicable cryptological systems. Full knowledge of keying devices and materials. With use of a guide, keyed all systems without error.

5.5.4.2. Q-. Familiar with applicable cryptological systems, keying devices and materials. With use of guide, keyed most systems with minor error.

5.5.4.3. U. Lacked knowledge of applicable cryptological systems, keying devices or keying materials. Failed to key systems without error.

5.5.5. **Area 212. Low Level Navigation.**

5.5.5.1. Q. Assisted the pilot with navigation through proper use/interpretation of systems. Maintained position awareness, terrain clearance, threat awareness/avoidance, Time-over-Targets (TOT), and terminal objective SA by identifying navigational cues. Safely cleared the aircraft as briefed/required.

5.5.5.2. Q-. Assisted the pilot with navigation through proper use/interpretation of systems with minor errors. Maintained position awareness, terrain clearance, threat awareness/avoidance, Time-over-Targets (TOT), and terminal objective SA by identifying navigational cues with some deviations. Safely cleared the aircraft as briefed/required.

5.5.5.3. U. Major navigation deviations which affected safe/effective mission accomplishment. Did not use navigational cues to maneuver the aircraft. Did not safely clear the aircraft as briefed/required.

5.5.6. **Area 213. Formation Procedures.**

5.5.6.1. Q. Thorough knowledge of formation procedures IAW technical orders and AFIs. Provided accurate and timely calls to keep crew informed of wingman's position through use of aircraft systems (e.g., Distance Measuring Equipment, FLIR, RADAR).

5.5.6.2. Q-. Limited knowledge of formation procedures or use of aircraft systems. Mission accomplishment not degraded.

5.5.6.3. U. Lacked ability to use aircraft systems to keep crew informed of wingman's position to a point of mission degradation or safety.

5.5.7. **Area 214. TF/TA Procedures.**

5.5.7.1. Q. Demonstrated knowledge of TF/TA by integrating radar with digital map display and other sensors to execute a successful low-level mission. Acknowledged and managed radar notifications (e.g., warnings and enunciators) and provided accurate calls in a timely manner. Recognized and rectified system failures or altered method of low level using other sensors as appropriate.

5.5.7.2. Q-. Minor errors in TF/TA procedures or employment. Demonstrated limited knowledge of radar capabilities or procedures; slow to recognize system failures and take appropriate action. Used other aircraft sensors with minor errors or deviations. Mission accomplishment not degraded.

5.5.7.3. U. Failed to employ TF/TA radar to ensure mission accomplishment. Misunderstood radar capabilities, failed to acknowledge radar notifications or failed to fully integrate/interpret all sensors while flying the profile.

5.5.8. **Area 215. Degraded Systems.**

5.5.8.1. Q. Demonstrated the ability to correctly identify, troubleshoot, and apply corrective measures to aircraft systems malfunctions. Knowledgeable of operational restrictions or limitations associated with degraded systems.

5.5.8.2. Q-. Minor errors in identifying, troubleshooting, or applying corrective actions to system malfunctions which would not degrade mission accomplishment.

5.5.8.3. U. Failed to correctly identify, troubleshoot, or apply correct measures to systems malfunctions. Mission accomplishment would have been severely impacted.

5.5.9. **Area 216. Infil/Exfil Procedures**

5.5.9.1. Q. Followed/explained proper procedures for infiltration/ex-filtration operations. Thoroughly familiar with proper light signals/communications and emergency procedures.

5.5.9.2. Q-. Difficulty following/explaining proper procedures for infiltration/ex-filtration operations. Limited knowledge of proper light signals/communications and emergency procedures that did not degrade mission effectiveness.

5.5.9.3. U. Did not follow/explain proper procedures for NVG infiltration/ex-filtration operations. Did not use proper light signals/communications and emergency procedures. Failed to use or understand proper light signals/communications and emergency procedures.

5.5.10. **Area 217. Alternate Insertion/Extraction Procedures.**

5.5.10.1. Q. Performed hoist and fast rope operations IAW the flight manual and other published directives. Aware of computed power requirements/limitations. Good knowledge of all hoist/fast rope limitations and malfunctions. Provided clear and concise direction during the approach, insertion, pickup and departure. Promptly advised the pilot flying of minor drift tendencies. Actively scanned/cleared the aircraft during the AIE procedures. Equipment malfunctions were dealt with effectively while maintaining situational awareness and crew coordination.

5.5.10.2. Q-. Same as Q above except for minor deviations which did not affect safety/mission accomplishment. Slow to advise the pilot flying of drift tendencies and/or slow to scan the area around the aircraft.

5.5.10.3. U. Major deviations to the procedures outlined in the flight manual and other published directives which affected safe/effective mission accomplishment. Unaware of the proper AIE methods or the emergency procedures associated with the method used. Poor knowledge of hoist/fast rope limitations and malfunctions. Failed to advise the pilot flying of drift tendencies and failed to scan the area around the aircraft. Equipment preparation was not accomplished in a timely manner or IAW published directives. Did not correct equipment malfunctions and failed to maintain situational awareness/crew coordination.

5.5.11. **Area 218. Air Refueling.**

5.5.11.1. Q. Was fully knowledgeable of all air refueling operations and procedures. Read correct checklists and performed required checklist items for each phase of flight. Maintained situational awareness and recognized all light signals. Closely monitored fuel management, aircraft systems/instruments and lighting. Assisted the pilot flying as briefed/required.

5.5.11.2. Q-. Was fully knowledgeable of all air refueling operations and procedures with some omissions. Minor deviations while reading checklists and performing checklist items for each phase of flight. Did not recognize/know all light signals and/or failed to closely monitor systems/instruments and lighting. Assistance to crew lacking.

5.5.11.3. U. Limited knowledge of air refueling operations and procedures. Failed to read and perform checklist items for each phase of flight. Failed to recognize/know any light signals. Failed to monitor fuel management, aircraft systems/instruments and lighting. Did not assist the crew as briefed/required.

5.5.12. **Area 219. Ramp Mounted Weapon System.**

5.5.12.1. Q. Demonstrated the ability to arm, de-arm, and safe the weapon in flight. Able to identify and clear weapons malfunctions in a timely manner IAW the appropriate operating manuals. Kept crew advised of other aircraft/teams/targets during employment.

5.5.12.2. Q-. Minor procedure errors and/or was slow to arm/de-arm/safe the weapon. Able to identify and clear weapons malfunctions with minor errors. Slow to advise crew of other aircraft/teams/targets during employment. Safety was never compromised.

5.5.12.3. U. Failed to arm, de-arm, or safe the weapon. Major errors in malfunction procedures. Failed to advise crew of other aircraft/team/survivor/target during employment. Compromised safety of flight.

5.5.13. **Area 220. Low Visibility Approach.**

5.5.13.1. Q. Assisted the pilot with navigation through proper use/interpretation of systems to maintain final approach altitude and ground speed to prebriefed location. Maintained terminal area objective SA by identifying navigational cues as briefed/required.

5.5.13.2. Q-. Slow to assist the pilot with navigation through proper use/interpretation of systems to maintain final approach altitude and ground speed to prebriefed location. Maintained terminal area objective SA by identifying navigational cues as briefed/required with errors. Marginal terminal area SA. Safety or mission accomplishment was not affected.

5.5.13.3. U. Did not assist pilots with navigation or inputs. Failed to maintain terminal area objective SA. Safety and mission accomplishment was compromised.

5.5.14. **Area 221. Systems Operations/Knowledge/Limitations.**

5.5.14.1. Q. Demonstrated/explained a complete knowledge of aircraft systems operations/limitations and proper procedural use of systems with and without reference to the flight manual or available aids.

5.5.14.2. Q-. Marginal knowledge of aircraft systems operations and limitations in some areas. Used individual technique instead of established procedures and was unaware of differences.

5.5.14.3. U. Unsatisfactory systems knowledge. Failed to demonstrate/explain the procedures for aircraft system operations.

5.5.15. **Area 222 Aircraft Defensive Systems.**

5.5.15.1. Q. Demonstrated a complete knowledge of aircraft defensive systems and operating limitations both with and without reference to the flight manual and/or available aids.

5.5.15.2. Q-. Limited knowledge of aircraft defensive systems operations and limitations in some areas. Used individual technique instead of procedures and was unaware of differences.

5.5.15.3. U. Unsatisfactory aircraft defensive systems knowledge. Failed to demonstrate or explain the procedures for aircraft systems operations with or without reference to the flight manual and/or available aids.

5.5.16. **Area 223. Night Water Hoist.**

5.5.16.1. Q. Performed hoist operations IAW the flight manual and other published directives. Aware of computed power requirements/limitations. Good knowledge of all hoist limitations and malfunctions. Provided clear and concise direction during the approach, insertion, pickup and departure. Advised the pilot flying promptly of minor drift tendencies. Actively scanned/cleared the aircraft during the AIE procedures. Equipment malfunctions were dealt with effectively while maintaining situational awareness and crew coordination.

5.5.16.2. Q-. Same as Q above except for minor deviations which did not affect safety/mission accomplishment. Slow to advise the pilot flying of drift tendencies and/or slow to scan the area around the aircraft.

5.5.16.3. U. Major deviations to the procedures outlined in the flight manual and other published directives which affected safe/effective mission accomplishment. Unaware of the proper AIE methods or the emergency procedures associated with the method used. Poor knowledge of hoist limitations and malfunctions. Failed to advise the pilot flying of drift tendencies and failed to scan the area around the aircraft. Equipment preparation was not accomplished in a timely manner or IAW published directives. Did not correct equipment malfunctions and failed to maintain situational awareness/crew coordination.

5.5.17. **Areas 224-249 reserved for future use.**

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Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 11-202V2, *Aircrew Standardization/Evaluation Program,* 13 September 2010

AFI 11-202V3, *General Flight Rules,* 10 August 2016

AFI 11-215, *USAF Flight Manuals Program (FMP),* 22 December 2008

AFI 11-218, *Aircraft Operations and Movement on the Ground,* 28 October 2011

AFI 11-290, *Cockpit/Crew Resource Management Training Program,* 15 October 2012

AFI 11-2CV-22V1, *CV-22 Aircrew Training,* 17 April 2015

AFMAN 33-363, *Management of Records,* 1 March 2008

AFPD 11-2, *Aircraft Rules and Procedures,* 14 January 2005

AFPD 11-4, *Aviation Service,* 1 September 2004

AFI 11-2CV-22V3, *CV-22 Operations Procedures,* 1 May 2017

AFTTP 3-3, *CV-22 Combat Aircraft Fundamentals,* 16 November 2015

Adopted Forms

AFSOC Form 48, Flight Evaluation Form

AF Form 8/8a, Certificate of Aircrew Qualification

AF 09138047, Recommendation for Change of Publication

AF Form 4031, CRM Skills Criteria Training/ Evaluation

AF Form 942 Record of Evaluation

AFTO Form 781, ARMS Aircrew/Mission Flight Data Recorder

Abbreviations and Acronyms

AIRCAT—Automated Integrated Radar Control for Air Traffic

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFSOC—Air Force Special Operations Command

AFTTP—Air Force Tactics Techniques and Procedures (AFSOCH 11-222 Combat Aircraft Fundamentals CV-22)

AIE—Alternate Insertion/Extraction

ARMS—Aviation Resource Management Systems

ASR—Airport Surveillance Radar

ATD—Aircrew Training Device

BAQ—Basic Aircraft Qualified

CRM—Cockpit/Crew Resource Management

DSO—Direct Support Operator

EPE—Emergency Procedures Evaluation

EP—Evaluator Pilot or Emergency Procedure

FDP—Flight Director Panel/Flight Duty Period

FE—Flight Engineer

FF—First Flight Engineer

FLIR—Forward Looking Infrared

FP—First Pilot

FTU—Flying Training Unit

IAW—In Accordance with

ILS—Instrument Landing System

INSTM—Instrument

LOC—Localizer

LVA—Low Visibility Approach

MAJCOM—Major Command

MAP—Missed Approach Point

MDA—Minimum Decision Altitude

MDS—Mission Design Series

MF—Mission Flight Engineer

MP—Mission Pilot

MSA—Minimum Safe Altitude

MSN—Mission

NAS—National Airspace System

NAVAID—Navigational Aid

NM—Nautical Mile

NVG—Night Vision Goggle

OPR—Office of Primary Responsibility

PAR—Precision Approach Radar

QUAL—Qualification

RQ—Requal

SA—Situational Awareness

SME—Special Mission Event

STAN/EVAL—Standardization/Evaluation

STOL—Short Takeoff and Landing

TA—Terrain Avoidance

TACAN—Tactical Air Navigation

TF—Terrain Following

TF/TA—Terrain Following/Terrain Avoidance

TOLD—Takeoff and Landing Data

TOT—Time Over Target

TTP—Tactics, Techniques, and Procedures

VDP—Visual Descent Point

VFR—Visual Flight Rules

VOR—Very High Frequency Omni Directional Range Station

W&B—Weight and Balance

Terms

Air Refueling—Airborne fuel on-load by receiver aircraft.

Boldface—Refers to Emergency Action Procedures that are Bolded in NATOPS and must be memorized.

Deviation—Performing an action not in sequence with current procedures, directives, or regulations. Performing action(s) out of sequence due to unusual or extenuating circumstances is not considered a deviation. In some cases, momentary deviations may be acceptable; however, cumulative momentary deviations will be considered in determining the overall qualification level.

Error—Departure from standard procedures. Performing wrong actions or recording incorrect information.

Low Level Operations—conducted below 3,000 feet AGL.

Minimum Safe Altitude (MSA)—MSA is an intermediate altitude which will provide terrain clearance in VMC or Instrument Meteorological Conditions.

Minor—Did not detract from mission accomplishment, adversely affect use of equipment, or violate safety.

Prerequisite—Requirements that must be completed prior to beginning an evaluation. Requires AF Form 8/8a documentation.

Recheck—A subsequent evaluation used to remedy a failed evaluation. Requires AF Form 8/8a documentation.

Requisites—Requirements such as exams, EPEs, Boldface/CAPs, etc., that have to be accomplished before an evaluation is considered complete. Requires AF Form 8/8a documentation.